# Albergheria–Ballarò, Palermo

*The El Niño heat deaths for Sicily have already happened, in summer 2026. The next ones will come in summer 2027, so every isolated older person in the quarter needs a named person calling them by late May.*

| REGION | EXCESS HEAT DEATHS | WINDOW | DOMINANT HAZARD |
|---|---|---|---|
| Sicily (GADM ITA.15_1), region level | 930 already past (June–September 2026); 37 forecast October 2026 – March 2027 | 6-month forecast, Oct 2026 – Mar 2027; summer 2027 is not covered | Summer heat with power cuts; flash floods this winter; drought less likely now the reservoirs have refilled |

**22/22 recommendations checked.** 9 confirmed, 13 corrected, 0 unverified, 0 contraindicated among the recommendations that remain. That covers the 8 ranked actions (2 confirmed, 6 corrected), the 5 "Do not do this" cards (3 confirmed, 2 corrected), and the 9 For you steps (4 confirmed, 5 corrected). One action the panel proposed, a cold room in the empty covered market, was dropped as unworkable. Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review. It is published for Sicily, not for the quarter, and counts heat only. 930 of Sicily's 970 projected deaths fell in June–September 2026, so this briefing is about summer 2027, which the tracker does not cover. Nothing in it depends on El Niño: the same heat risk would exist without it. After verification, the funding target fell from €181,000 (US$202,213) to €85,000 (US$94,962), down 53%, because the first figure counted every

EXECUTIVE SUMMARY — AT A GLANCE




38.11°N 13.36°E, Italy · Natural Earth

- **The risk.** Summer heat with power cuts: Palermo's deaths ran 30% above normal in July 2023 and 19% above in July 2026, mostly among the very old.
- **For you.** By 31 March, ask your GP whether you, or an older relative, are on the health authority's heat frailty list. It costs nothing.
- **The money.** €16,100–181,000 (US$17,987–202,213); a named caller for each isolated older person buys the most.
- **Do not.** Do not tell an older person a fan will keep them safe: in a 2024 trial at 36 °C, it did not lower their body temperature.*


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* Suggested formulation: check with a health worker before sharing.

resident over 75 rather than those living alone, and a cash line duplicated an automatic national bill discount. The Climate Impact Lab Adaptation Inventory loads in the browser and returned no entries to the research agents, so no precedent could be confirmed there.

FOR YOU, AS A RESIDENT

9 STEPS

In Albergheria the danger is a hot July night in an upper-floor flat with the power off, and the people at risk are mostly over 85 and living alone. Most of what protects them costs little and has to be set up in winter and spring, before the first heatwave. The ranked actions below are written for the city and its partners. These are the ones you can take yourself.

### Ask your GP (medico di base) whether you, or an older relative, are on the health authority's heat frailty list

PARTIALLY CONFIRMED | FROM ACTION 1

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 31 March 2027 | none | people over 75, and anyone chronically ill or housebound |

Palermo's 2026 heat plan has the health authority (ASP) send GPs a list of patients at high risk in heat, for the GP to check. There is no public way to put yourself on it, so ask the GP directly, and say if the person lives alone or uses oxygen or dialysis at home.

### Name two older neighbours who live alone, and agree to check on them every day there is an orange or red heat alert

PARTIALLY CONFIRMED | FROM ACTION 1

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 31 May 2027 | none | your neighbours |

In July 2026 Palermo's excess deaths were concentrated among people over 85. In Rome, a volunteer programme that kept in touch with isolated older people was associated with fewer than half the expected heat deaths among those over 80. That study was observational, but the step costs nothing. A knock at the door from someone they know is what most of them will answer.

* Suggested formulation: check with a health worker before sharing.

### Follow the daily heat bulletin from mid-May, and keep 1500 and 112 to hand

CONFIRMED · FROM ACTION 2

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| from late May 2027 | none | the household |

The Ministry of Health issues a heat bulletin for Palermo on weekdays from late May to September, on salute.gov.it and the "Caldo e Salute" app, with a three-day forecast that covers weekends. Level 3 means three or more days of dangerous heat in a row. The national heat helpline is 1500 in summer, and 112 is the emergency number.

### Keep the flat cool without relying on the grid: shutters closed by day, windows open at night, cool showers, a wet cloth *

CONFIRMED · ROLE-SPECIFIC

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| from June 2027 | none | everyone at home, including during a power cut |

On ordinary humid summer days (32–35 °C), a fan with wet skin is fine as one part of keeping cool. On scirocco days above about 38 °C, do not rely on a fan: take an older person to a cooled room, use cool showers, and check on them. See "Do not do this".*

### If anyone at home uses electric oxygen or dialysis, apply to the Comune for the electricity bonus for physical hardship, and plan for an outage of two days

PARTIALLY CONFIRMED · FROM ACTION 3

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 30 April 2027 | none to apply | people on home medical equipment |

The bonus per disagio fisico is recorded against your electricity supply point by the distributor, e-Distribuzione, which makes the household visible to it. It does not guarantee faster repair. In July 2023 some Palermo streets were without power for up to 42 hours, so agree a backup plan with the doctor who prescribed the equipment.*

* Suggested formulation: check with a health worker before sharing.

### If you hold a Carta Dedicata a te, make your first purchase by 16 December 2026

CONFIRMED · FROM ACTION 6

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 16 December 2026 | none; it protects €500 (US$559) | household food budget |

The 2026 card is loaded from 4 November. If it is not used once by 16 December, this year's €500 is lost. Palermo has 18,447 holders, the third-highest number in Italy. Eligible households (ISEE up to €15,000) do not apply: the card is issued automatically.

### File an ISEE declaration at a CAF tax-assistance centre by 31 March 2027, and keep your electricity bills

CONFIRMED · FROM ACTION 6

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 31 March 2027 | none at a CAF | the cost of running cooling |

If your ISEE is €9,796 (US$10,944) or less (€20,000 with four or more children), the social bonus on electricity, water, and gas is applied to your bills automatically. If a power cut lasts more than about eight hours, the bill should show an automatic credit of about €34.50 (US$38.54), plus about €17.25 (US$19.27) for each further four hours; if it does not, claim it.

### If you plan to buy air conditioning, buy a heat-pump unit for one bedroom before 31 December 2026

PARTIALLY CONFIRMED · ROLE-SPECIFIC

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 31 December 2026 | about €500–1,000 (US$559–1,117) installed, a national range; 50% back over 10 years for spending in 2026, 36% in 2027 | the most vulnerable person in the household |

In 2021, about one Sicilian household in three had no air conditioning. The tax deduction for a heat pump in a main home falls from 50% to 36% on 1 January 2027, so buying this year costs less. Do not make it the only plan: the local cables fail on the hottest nights, which is when it would be needed.

* Suggested formulation: check with a health worker before sharing.

### Before June, put together a blackout kit: a rechargeable fan, a power bank, a cool box with ice packs, and two weeks of food that needs no cooking

PARTIALLY CONFIRMED | ROLE-SPECIFIC

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 31 May 2027 | about €75 (US$84), an unverified estimate | food, medicines that need cooling, and a working phone |

In July 2023, heat, fires, blackouts, and water cuts hit Palermo in the same week, and outages recurred in July 2026, including at care homes in Libertà. If anyone uses insulin, ask the pharmacist how to keep it safe without a fridge. Prices are estimates from the panel, not checked locally.*

## What to ask of the people in charge

- Ask your ASP district, or the Comune's social services, whether the people on the frailty list in Albergheria have a named person who calls them on alert days (action 1).
- Ask the Comune's civil protection office where the quarter's cooled room will be, and whether it opens at level 2 or only after the crisis centre is set up (action 4).
- If you trade at Ballarò, ask the Comune whether a market heat ordinance will cover stallholders in 2027, since the regional ordinance covers employees only (action 5).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak, here the first heatwaves of summer 2027. Backing lenses are shown per

* Suggested formulation: check with a health worker before sharing.

action; an action several lenses reached independently is the stronger bet.

## 1. Give every person on the frailty list in Albergheria a named caller by 25 May, and add oxygen and dialysis users to the list

PARTIALLY CONFIRMED · PUBLIC HEALTH · EMERGENCY MANAGER · FRONTLINE · HISTORIAN · PHILANTHROPIST · SKEPTIC · ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| ASP Palermo and GPs, with the Comune's social services and local volunteers | list confirmed by 31 Mar 2027; calls live by 25 May 2027 | €45,000 (US$50,274) for 250 people at €180 (US$201) each, an estimate | deaths indoors among the very old and isolated, early in the season |

The list already exists. Palermo's 2026 heat plan has the regional health department send the ASP a list of high-risk residents, each with their GP, for the GP to check, and the Comune's social services receive it. What the plan lacks is someone calling those people routinely: checks start only when the crisis centre opens on level-3 days, and they respond to calls. Volunteers from the quarter's parishes and social centres (San Saverio, Santa Chiara, Sant'Egidio) would phone or visit each person on every orange and red day. In Rome, Sant'Egidio's programme for isolated people over 80 was associated with heat mortality of less than half the expected rate (IRR 0.44, 95% CI 0.32–0.60) at about €81 (US$90) a person a year.

Transferability: good. The evidence is Italian and urban, but observational, so it shows an association. Early-season heatwaves kill disproportionately because people have not adapted, so a scheme that starts at the first red alert in July is already late. The number of people over 75 living alone in the quarter is unknown; 250 is an estimate.

## 2. Relay the heat bulletin to people who never see it, in their languages and at weekends

CONFIRMED · FRONTLINE · PHILANTHROPIST · PUBLIC HEALTH · HISTORIAN · ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Comune di Palermo civil protection, with volunteers and migrant associations | voice notes recorded by 15 May 2027; relay from the first bulletin | €8,000 (US$8,938), an estimate | people not knowing that a dangerous day is coming |

* Suggested formulation: check with a health worker before sharing.

Palermo is one of 27 Italian cities with a Ministry of Health heat bulletin, issued at 11:00 on weekdays with a 72-hour forecast. Volunteers record 60-second voice notes in Italian, Bengali, Arabic, and English on the warning signs, 112, and where the cooled room is, and pass them on through shopkeepers, Moltivolti, and parishes by WhatsApp, using the 72-hour forecast to cover weekends. Bangladeshis are the city's largest foreign community (5,932 of 25,135 foreign residents in 2024), and many are in Ballarò. Across 16 Italian cities, the effect of extreme heat on deaths of older people fell from +36.7% to +13.3% after prevention plans began, although the authors also credit changing temperature patterns.

Transferability: good. This adds a last mile to a system Italy already pays for. In Philadelphia, a warning system costing about €188,000 (US$210,000) was estimated to save 117 lives over four years.

## 3. Before summer, make the households on medical equipment visible to the grid, and agree a restoration order for the quarter

PARTIALLY CONFIRMED | EMERGENCY MANAGER · SKEPTIC | ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| **Comune civil protection, with e-Distribuzione and the Comune's social services** | **30 Apr 2027** | **staff time** | **deaths when a heat blackout cuts oxygen, dialysis, or cooling** |

Households using electric oxygen or dialysis should be enrolled in the electricity bonus for physical hardship, which the distributor records against their supply point. No rule gives them faster repair, so the civil protection office should then agree with e-Distribuzione a restoration order for the feeders serving the quarter, with care homes and those households first, and a protocol for insulin and other medicines that must stay cool. In July 2023 some areas were without power for up to 42 hours. In July 2026, care homes in Libertà lost power, Arenella was cut off for several days, and the Civico hospital's emergency department lost power.

* Suggested formulation: check with a health worker before sharing.

Transferability: this is a local failure. Enel attributes the summer outages to cables that cannot shed heat, not to demand alone, so the cuts will recur until the network is upgraded.

## 4. Open one cooled room with backup power in the quarter, at level 2, from 1 June

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Comune civil protection, with a host centre in the quarter | 1 Jun 2027 | €17,000 (US$18,992): two heat-pump units, a battery or generator, and 90 days of staffing | having nowhere cool to go when heat and a blackout coincide |

The 2026 plan promises air-conditioned reception areas only once the crisis centre is set up after a third level-3 day, and names no sites or hours. Otherwise it tells people to go to cinemas, shopping centres, and libraries. On 24 July 2026, the city's libraries suspended public service because their air conditioning was not working. A room open from 11:00 to 19:00 on every orange and red day, with power that survives a cut, fills that gap. San Saverio or the Salesian centre at Santa Chiara are candidates, but San Saverio's building was under threat in May–June 2026, so funding depends on a written agreement with the owner.

Transferability: cooling centres are widely recommended but their effect on deaths is unproven, and under-use is well documented. Pair this room with the named callers in action 1, who can bring people to it.

## 5. Make the heat rules reach Ballarò's market and riders: the Region reissues its ordinance, and the Mayor covers stallholders

PARTIALLY CONFIRMED | ECONOMIST · FRONTLINE | ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Mayor of Palermo, with the Regione Siciliana | 1 Jun 2027 | a council decision; the lost awning fees are unquantified | heatstroke among older stallholders and riders working through midday |

* Suggested formulation: check with a health worker before sharing.

The Region's 2026 ordinance banned outdoor work from 12:30 to 16:00 in construction, agriculture, quarries, and logistics, including delivery riders, wherever the INAIL Worklimate map showed "high" risk at noon. It lapsed on 30 September and must be reissued for 2027, as it has been each year since 2024. It binds employers only, so it does not cover the self-employed stallholders of Ballarò. A mayoral ordinance could shift trading hours on high-risk days and waive the street-occupancy fee for shade awnings. Unions say the regional ban has not been inspected, so a card explaining riders' rights helps.

Transferability: this is local. The Comune is under a long-term debt-recovery plan, so any fee waiver is a choice to give up revenue.

## 6. Help residents claim money they are already owed: the Carta Dedicata a te by 16 December, and the electricity bonus by March

CONFIRMED · ECONOMIST · FOOD SUPPLY CHAIN · ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Comune di Palermo social services, with CAF centres and parishes | 16 Dec 2026 and 31 Mar 2027 | about €10,000 (US$11,172) for outreach, an estimate | households losing €500 of food money, and not affording to run cooling |

The 2026 Carta Dedicata a te pays €500 (US$559) to households with an ISEE up to €15,000; the first purchase must be made by 16 December or this year's money is lost. The social bonus on electricity applies automatically to households with an ISEE of €9,796 (US$10,944) or less, but only once a declaration (DSU) has been filed. Volunteers and parishes go door to door in the quarter, and CAF centres file declarations. Food prices are rising faster for the poorest: Italian inflation was 4.2% in September 2026, with unprocessed food up 5.5%, energy up 22.6%, and prices for low-spending households up 4.4% against 2.9% for the highest spenders.

* Suggested formulation: check with a health worker before sharing.

Transferability: the schemes exist and pay automatically once a household is registered; the barrier is paperwork, which is what volunteers can fix.

## 7. This winter, publish the underpass closure rule and clear the drains along the old riverbeds

PARTIALLY CONFIRMED | HISTORIAN | ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Comune civil protection, with AMAP | now, through the September–January storm season | within existing maintenance budgets; not costed | people trapped in flooded underpasses and streets |

On 15 July 2020 the underpasses on viale Regione Siciliana flooded with no regional alert issued. No one died, but courts later found the Comune liable for failing to maintain the drains. Seven flood sensors have since been installed in those underpasses, and they are closed in storms in practice, but no formal closure protocol has been published. Ballarò itself sits on the bed of the Kemonia river, which flooded the quarter in 1557 and 1931.

Transferability: local. This is not an El Niño effect; it is the hazard of this season.

## 8. If water rationing returns in 2027, publish the schedule only on official channels by 31 May and put listed people on the tanker priority list

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NOT CHECKED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| AMAP | 31 May 2027, if needed | within AMAP's existing budget | older people without water during a heatwave; confusion from fake schedules |

The 2024 drought brought rationing from July 2024, with weekly 24-hour cuts by district from October; the historic centre was largely exempt. Rationing ended on 16 February 2026, after rain refilled the reservoirs: on 1 September 2026 Rosamarina held 56 million m³ and Poma 46 million m³, against about 20 million each a year earlier. In 2024, AMAP had to deny a fake rationing calendar spread on social media and chats.

* Suggested formulation: check with a health worker before sharing.

Transferability: rotation is unlikely in 2027 given the reservoirs. Every past water crisis here ended because of the weather, not because of repairs, and the leaks remain.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

## RACI CHART

**R** = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

| Action | ASP Palermo and GPs | Comune social services | Mayor and Comune civil protection | e-Distribuzione | Regione Siciliana | AMAP | Parishes and volunteers |
|---|---|---|---|---|---|---|---|
| 1. Named callers for the frailty list | **A** | **R** | **C** | | | | **R** |
| 2. Bulletin relay | **C** | **I** | **A** | | | | **R** |
| 3. Grid visibility and restoration order | **C** | **R** | **A/R** | **R** | | | **I** |
| 4. Cooled room with backup power | **C** | **C** | **A** | **I** | | | **R** |
| 5. Market and rider heat rules | **I** | | **A/R** | | **R** | | **I** |
| 6. Claim money owed | | **A/R** | | | | | **R** |
| 7. Underpasses and drains | | | **A** | | **I** | **R** | |
| 8. Water schedule, if needed | | **C** | **C** | | **C** | **A/R** | **I** |

What the chart shows: the Mayor and the civil protection office are accountable for five of the eight actions, and parishes and volunteers do the work in five. Rows 3, 5, 7, and 8 need decisions, not new money.

WHERE MONEY DOES THE MOST

## FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per euro. The envelope is bounded by what this place can spend before the peak, not by what it needs.

| FLOOR | TARGET | CEILING | LARGEST UNCERTAINTY |
|---|---|---|---|
| **€16,100 (US$17,987)** | **€85,000 (US$94,962)** | **€181,000 (US$202,213)** | **How many people over 75 live alone in the quarter; nobody has counted** |

Floor: the bulletin relay (€8,000) plus named callers for the 100 frailest at Sant'Egidio's Rome cost of €81 each (€8,100), €16,100 in all. Target: the relay (€8,000), callers for 250 people at €180 each, allowing for phones and visits (€45,000), the cooled room (€17,000, only with a tenure agreement), and €150 each in electricity cash for 100 households that cannot obtain an ISEE (€15,000), €85,000 in all. Ceiling: the relay, callers for 700 people (€126,000), the room, and cash for 200 households (€30,000), €181,000 in all. Above that, volunteer-run groups cannot spend more before June. The panel proposed €181,000 as the target; it fell by 53% because 700 is roughly every resident over 75, not those living alone (estimated at 200–280), and electricity cash for households with an ISEE duplicates the automatic national bonus.

* Suggested formulation: check with a health worker before sharing.

| RECIPIENT | USE | AMOUNT | MECHANISM | BASIS | VERDICT |
|---|---|---|---|---|---|
| Volunteer network in the quarter (San Saverio, Santa Chiara) with migrant associations such as Moltivolti | Action 2: multilingual bulletin relay | €8,000 (US$8,938) | direct grant | An estimate; Philadelphia's warning system, about €188,000 (US$210,000) and an estimated 117 lives (Ebi 2004), as an analogue | Partially confirmed |
| Comunità di Sant'Egidio Palermo, or the quarter's volunteers if it will not run the programme here | Action 1: named callers for isolated older people | €8,100–126,000 (US$9,049–140,767); target €45,000 (US$50,274) | top-up | €81 (US$90) a person a year, Sant'Egidio's Rome figure; €180 allows for phones and visits; IRR 0.44 in Rome (Orlando 2021) | Partially confirmed |
| Host centre in Albergheria (San Saverio or Santa Chiara) | Action 4: cooled room with backup power | €17,000 (US$18,992) | prepositioned stock | Two heat-pump units €3,000, battery or generator €5,000, 90 days' staffing €9,000; no published cost per life. Conditional on written tenure | Partially confirmed |
| Caritas or Centro Astalli, for households outside the ISEE system | Action 6: electricity cash where the national bonus cannot reach | €15,000–30,000 (US$16,758–33,516) | cash transfer | €150 a household; the weakest evidence of the four | Partially confirmed |

The funder's one thing: Albergheria has €12.4 million (US$13.9 million) from the national recovery plan for building works, and a regional fund of €1 million (US$1.12 million) a year shared with three other quarters for schools and advice desks. Almost nothing pays for the people inside those buildings during a heatwave. The cheapest life saved here is a phone call to someone who would otherwise be alone on a red day.

* Suggested formulation: check with a health worker before sharing.

# Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

## Do not tell an older person that a fan will keep them safe *

CONTRAINDICATED CONFIRMED

In a 2024 randomised trial (JAMA), adults aged 65–85 spent eight hours at 36 °C and 45% humidity: a fan did not lower their peak body temperature. Above about 38 °C, the temperature of a scirocco day, a fan blows hot air and can add heat.

**Instead:** on humid days, a fan with wet skin as one part of keeping cool; on scirocco days, a cooled room, cool showers, and someone checking in.

## Do not publish a map of "climate shelters" as the main heat measure

CONTRAINDICATED CONFIRMED

Palermo's 2026 plan promises air-conditioned reception areas but names no sites, and opens them only once the crisis centre is set up. On 24 July 2026 the city's libraries, one of the places people were told to go, suspended service because their air conditioning had failed. The people most at risk do not leave the house.

* Suggested formulation: check with a health worker before sharing.

**Instead:** a named caller for each person on the frailty list (action 1), and one cooled room with backup power that the callers can bring people to (action 4).

### Do not hand out air conditioners as the solution

CONTRAINDICATED CORRECTED

The local network fails in heat: in July 2023 some streets were without power for up to 42 hours, and in July 2026 outages hit care homes, a hospital emergency department, and whole districts. Enel attributes this to cables that cannot shed heat. An air conditioner does nothing in a blackout, and it costs money to run.

**Instead:** shutters and night ventilation, the grid measures in action 3, and a cooled room with its own backup power.*

### Do not plan for summer 2027 around El Niño

CONTRAINDICATED CONFIRMED

The heat tracker's window ends in March 2027 and does not cover summer 2027. El Niño's signal in Mediterranean summers is weak and inconsistent in the research, and World Weather Attribution found that the 2024 Sicilian drought was driven by heat, with no significant change in rainfall. Waiting for an El Niño forecast, or calling the next drought an "El Niño drought," points attention at the wrong cause.

**Instead:** act on the Ministry's heat bulletin levels each summer, and press AMAP to fix the network's leaks.

### Do not trust a rationing calendar forwarded on social media or chats

CONTRAINDICATED CORRECTED

In 2024 a fake AMAP rationing calendar circulated on social media and chats, and AMAP had to deny it publicly. People who store water or stay home on the wrong days are worse off.

**Instead:** check AMAP's own website and channels, and pass on only the official schedule.

* Suggested formulation: check with a health worker before sharing.

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| 1557 | Kemonia river flood, 27 September; the wave came down the riverbed through Ballarò | Not established; "there was talk of 2,000 dead"; about 1,000 houses destroyed | The quarter still sits on the old riverbed |
| 1931 | Flood, 20–23 February; about 435 mm of rain; the Kemonia and Papireto overflowed | Not established; 9 to 15 drownings were reported | The Passo di Rigano channel was built afterwards |
| 2003 | European heatwave | No Palermo figure; deaths over 75 in southern Italian cities rose 26% | Italy's heat warning system was set up; Palermo is one of its 27 cities |
| 2020 | Flash flood, 15 July; viale Regione Siciliana underpasses flooded with no alert issued | No deaths | Courts found the Comune liable for drain maintenance; seven flood sensors installed |
| 2023 | Heat and fires, 24–25 July: 47.0 °C at the observatory, a record; 4,314 ha burned in the comune; airport closed; outages up to 42 hours | 3 deaths in the province on 25 July; city deaths 30% above expected, 1 July – 4 August | Regional ban on midday outdoor work from 2024 |
| 2024–26 | Drought: national emergency declared 6 May 2024; AMAP rationing from 22 July 2024 | None recorded | Rationing ended 16 February 2026, when rain refilled the reservoirs; leaks unfixed |
| 2026 | July heat: blackouts in Libertà (1 July) and Arenella (from 21 July); libraries closed on 24 July | 533 deaths against 448 expected in July (+19%), mostly over 85; Sicily 930 excess heat deaths June–September (tracker) | Not yet known |

The recurrence nobody has noticed: Ballarò is built on the bed of the Kemonia, and every water crisis Palermo has had, too much water or too little, has ended because of the weather, not because of new works. The 2026 drought ended with rain while the network's leaks remained. The heat crises follow the same pattern: each summer the plan switches on in July, and each summer the excess deaths fall among the very old in the first hot weeks.

* Suggested formulation: check with a health worker before sharing.

## WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **This is an El Niño briefing** versus **El Niño adds nothing here**. The Skeptic argued that the El Niño frame is irrelevant to Palermo's summer and could distract. The verifier agreed, and this briefing adopts that view: its actions would be the same without El Niño.
- **Buy air conditioning** versus **the grid fails when it is needed**. The Public Health and Philanthropist lenses recommend cooling one room for people who have none. The Skeptic warned against handing out units onto failing cables. The briefing leaves air conditioning as a household choice and spends public money on a room with its own backup power.
- **Insure market traders against hot days** versus **no such product exists**. The Economist proposed a parametric income pilot for Ballarò stallholders, paying on Worklimate "high" days. No such product is sold in Italy; it is a proposal, unfunded.

## WHAT NOBODY COVERED — BLIND SPOT

The frailty list is built from GP records, so it cannot see anyone without a GP or a registered residence, including undocumented migrants and some homeless people in and around Ballarò. No lens counted them or found who reaches them in a heatwave. Centro Astalli and Caritas are the likely channels. Nobody has counted how many people over 75 live alone in Albergheria either; ISTAT census-section data would answer that.

## HOW TO READ THIS — METHOD

**The panel is author-built.** Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are published at region level, here the whole of Sicily. They are not quarter figures. The El Niño projection covers *heat only* and ends in March 2027; summer 2027, flood, and drought are carried here by the historical record rather than by a published number.

## HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** This briefing was built on Angus Shaw's initiative, not from a request, as the first for Italy, the next country on the heat tracker's list without one. Sicily has Italy's second-highest projected total (970, after Lombardy's 1,400) and a higher rate per head. Albergheria–Ballarò,

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around the Ballarò market in Palermo's historic centre (Circoscrizione I), was chosen as a poor quarter with older residents, migrant communities, and a market workforce. The quarter's population is not published; Circoscrizione I had 25,516 residents in 2021. Mortality figures exist only at region level; quarter detail is used for hazard exposure and action design, not for death counts.

- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Sicily (GADM ITA.15_1), 930 additional heat deaths June–September 2026 (June 64, July 300, August 490, September 82), 37 forecast October 2026 – March 2027 (October 1, November 3, December 7, January 10, February 11, March 6; the months sum to 38 because of rounding), 970 in total; Italy, 7,500, of which 7,200 fell in June–September; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. Because the tracker's deaths had already happened, the lenses were asked to plan for summer 2027.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in Italy, a named payer, a trigger, and whether they already exist. Food actions were checked for published price series and delivery capacity. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. Every event in the record was checked against its source and NOAA's El Niño index.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Could not be consulted: the inventory loads its entries in the browser and returned none to the research agents. No action is tagged with a precedent from it.

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- **Money.** Costs are shown in euros with US dollars in parentheses, at €1 = US$1.1172 (ECB reference rate, 30 October 2026).
- **Reader.** No role was given, so the For you section is written for a resident. Its steps were taken from the verified actions and from what the panel found about households here; steps not drawn from a ranked action were verified on the same three tests. The rest of the report is written for the city and its partners.
- **Health wording.** Passages where a wording error could hurt someone (cooling and fans for older people, medicines, and medical equipment in an outage) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number. Estimates are labelled as estimates. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in Albergheria was interviewed. Two verifiers disagreed on whether the July 2026 Palermo death figures (533 against 448 expected) appear in the cited Il Post article; they are given as reported there. The quarter's population, its number of older people living alone, and local prices for household items could not be found. One source on library closures returned an error to one verifier.

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References · Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); tracker · confirmed
Comune di Palermo, Piano Operativo Ondate di Calore, Estate 2026 · confirmed; ASP Palermo 2026 plan, La Sicilia · confirmed; Regione Siciliana, Linee guida ondate di calore · confirmed
Orlando et al., IJERPH 2021 · confirmed (ecological); Sant'Egidio cost, Viva gli Anziani press kit · confirmed; Sant'Egidio in Palermo, LiveSicilia · confirmed
Schifano et al., Environ Health 2012 · confirmed; de'Donato et al., Public Health 2018 · confirmed; Ebi et al., BAMS 2004 · confirmed (secondary tables); heat bulletin season, Quotidiano Sanità · confirmed; 1500 line, Dire · confirmed; 112 in Palermo, Balarm · confirmed
July 2026 mortality, Il Post · confirmed by one verifier, figures not found by another; 2023 mortality, BlogSicilia · confirmed; air conditioning in Sicily, ISTAT 2021 · confirmed
O'Connor et al., fans in older adults, JAMA 2024;332(20):1752, doi:10.1001/jama.2024.19457 · confirmed; Morris et al., Lancet Planet Health 2021 · confirmed
Outages: July 2023, Libertà 2026, Arenella 2026, Civico 2026, Enel on causes · confirmed; bonus per disagio fisico, ARERA · confirmed; outage compensation, Altroconsumo · confirmed (secondary)
Libraries closed, Balarm · confirmed by one verifier, error for another; cooling centres, GHHIN evidence summary · confirmed; San Saverio, Balarm and La Sicilia · confirmed; Santa Chiara, Balarm · confirmed; Moltivolti, italia.it · confirmed; foreign residents, Tuttitalia · confirmed


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Regional heat ordinance 2026, Regione Siciliana and extension · confirmed; enforcement, La Sicilia · confirmed; Comune finances, LiveSicilia · confirmed

Carta Dedicata a te 2026, QuiFinanza · confirmed; social bonus threshold, ARERA · confirmed; ISTAT, consumer prices, September 2026 · confirmed; tax deduction 2026–27, Lavori Pubblici · confirmed

Floods: 15 July 2020, no victims, court ruling, sensors, 1931, 1557 · confirmed

Heat and fires 2023: INAF, Il Post · confirmed

Water: rationing 2024, centre exempt, end of rationing, reservoirs, 1 Sep 2026, fake calendar · confirmed

El Niño and the Mediterranean: World Weather Attribution, Climate Dynamics 2013 · confirmed

Exposure data · Climate Impact Lab, 451,000 at Risk (superseded) · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Albergheria–Ballarò, Palermo. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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